MGTF 405 BUSINESS FORECASTING

Fall 2026, Master of Quantitative Finance

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COURSE OBJECTIVES

Forecasts of uncertain future outcomes or events form a key input into most business and investment decisions and so affect all areas of finance and business practice. Stock pickers and investment analysts need to forecast future returns on the stocks they consider investing in. Airlines need to forecast fuel prices to determine whether to hedge their exposure to future oil prices. Companies need to forecast both the expected value and volatility of their future cash flows to determine whether to use debt or equity in their financing decisions. The Federal Reserve decides current interest rates based on their predictions of future inflation and economic activity. Home builders need to plan their construction activities years in advance and so have to predict both house prices and demand for homes before committing millions of dollars into essentially irreversible investment projects. Pharmaceutical and IT firms are faced with both technological uncertainty (which products are feasible?) and uncertainties about future market size, competitors' behavior, and market shares. Predictions of these variables will affect which projects should be launched immediately, which ones should be canceled, and which ones should be postponed.

This course introduces students to quantitative methods for producing their own state-of-the-art forecasts based on information from sources such as past values of the variable(s) being predicted, surveys, market information and other relevant economic data. It also teaches course participants to become critical consumers of forecasts reported in the media and by professional forecasters. Are such forecasts of much value and, if so, how good is a forecaster's track record?

The course relies extensively on quantitative methods. Empirical methods such as time-series models, machine learning, regression analysis, nonlinear estimation and graphical tools lie at the core of the course and will be used extensively. Reflecting recent developments in the field, the course also introduces prediction markets and large language models (LLMs) as emerging tools and sources of information for forecasters — both as objects of study in their own right and as inputs that increasingly compete with, and complement, traditional statistical forecasts.

COURSE MATERIALS

Elliott, Graham, and Allan Timmermann, Economic Forecasting. Princeton University Press, 2016.

This book (referred to as ET below) provides a comprehensive coverage of a wide range of quantitative forecasting topics. The coverage sometimes becomes quite technical and on occasion we will use the book for background reading. A set of background papers can be downloaded from the web.

COURSE GRADES

- Class participation (10%)
- Assignments 1-3 (25%)
- Midterm exam (25%)
- Forecasting project (40%)

TEACHING ASSISTANT

The teaching assistant for this course is Luka Vulicevic. Luka will be available to answer questions in person and via e-mail during the week. His e-mail is lvulicevic@ucsd.edu .

PROFESSOR CONTACT INFORMATION

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STUDENTS WITH DISABILITIES

A student who has a disability or special need and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the faculty member.

The student must present the OSD letter of certification and OSD accommodation recommendation to the appropriate faculty member in order to initiate the request for accommodation in classes, examinations, or other academic program activities. No accommodations can be implemented retroactively.

Please visit the OSD Web site (osd.ucsd.edu) for further information or contact the Office for Students with Disabilities at (858) 534-4382 or fosorio@ucsd.edu.

ACADEMIC INTEGRITY

Integrity of scholarship is essential for an academic community. As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The University expects that both faculty and students will honor this principle and in so doing protect the validity of University intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind.

The complete UCSD Policy on Integrity of Scholarship can be viewed at: http://www-senate.ucsd.edu/manual/appendices/app2.htm#AP14

MGTF 405 COURSE OUTLINE

Week 1, September 28: Introducing the Forecasting Problem

- Course overview
  - Challenges facing forecasters
  - The forecasting process
  - Costs of making forecast errors: The loss function
  - Forecast horizon
  - Formal forecasting models
  - Desirable Properties of Forecasts
- The changing forecasting landscape: model-based forecasts, judgment, prediction markets, and LLMs as competing and complementary sources of predictions
- Simple Measures of Forecast Performance: MSE, MAE, directional accuracy
  - What does a good forecast look like? Graphical Analysis Tools
- Cold-start forecasting problems
- In-sample versus out-of-sample forecasts (backtesting)

Readings:

ET chapter 2, 15.1, 15.3.1, 16.1, 16.2.

Background readings

A. Timmermann, 2018, Forecasting Methods in Finance. Annual Review of Financial Economics, 10, 449-479.

Breitung, C., 2026, Text is all you need: Asset pricing without Returns. SSRN Working Paper.

Hong, H., J. D. Kubik, and A. Solomon, 2000, “Security Analysts’ Career Concerns and Herding of Earnings Forecasts.” Rand Journal of Economics, 31, 121-144.

Week 2, October 5: Time-series Forecasting Models: Learning from Past Information

- Time-series methods in forecasting
  - Mean, variance, autocorrelations, persistence
  - Covariance stationarity
  - White noise: the building block
  - Wold representation theorem
  - Moving Average and Autoregressive models
  - Chain rule of forecasting
  - Random Walk model with and without trend
  - Unit root tests
  - Seasonality and Trend

Readings:

ET chapter 7.1 - 7.4.

Week 3, October 12: Selecting a good Forecasting Model

- Linear Regression Model Used in Forecasting
  - OLS, Maximum Likelihood and GMM estimation
  - Data Transformation
- Model selection
  - Stepwise procedures, Information criteria, LASSO, Elastic Net, Bagging
- Machine Learning methods

Assignment 1 due

Readings:

ET Chapter 6.

Background readings

Goyal, A. and I. Welch, 2008, A Comprehensive Look at the Empirical Performance of Equity Premium Prediction. Review of Financial Studies 21(4), 1455-1508.

Kalfi, S. Y., A. Timmermann, and T van der Zwan, 2026, Overhyped? Can ML Models Reliably Predict Stock Returns? Working paper, UCSD.

Week 4, October 19: Unobserved variables, Machine Learning, Nonlinear Forecasting Models

- Machine learning and nonlinear forecasting models (cont.)
- Foundation Models — Time-series foundation models (TSLMs, e.g. TimeGPT-style architectures)
- Kalman filter – unobserved components models
- Markov switching models
- Exponential Smoothing

Readings:

ET chapter 7.5, 8.3, 8.5, Appendix A1 and A2 on the Kalman filter

Background readings

Ang, A. and A. Timmermann, 2012, Regime Changes and Financial Markets. Annual Review of Financial Economics 4: 313-337.

Aruoba, S.B, F.X. Diebold, and C. Scotti, 2009, Real-time measurement of business conditions. Journal of Business and Economic Statistics 27, 417-427.

Gu, S., B. Kelly, and D. Xiu, 2020, Empirical Asset Pricing via Machine Learning. Review of Financial Studies 33, 2223-2273.

Rahimikia, E., Ni, Hao, and Weiguan Wang, 2026, Re(Visiting) Time Series Foundation Models in Finance. Working paper, University College London.

Carriero, A., D. Pettenuzzo, and S. Shekhar, 2025, Macroeconomic Forecasting with Large Language Models. arXiv:2407.00890.

Week 5, October 26: Multivariate forecasting models

- Vector Auto Regressions (VARs)

2-hour midterm exam.

Readings:

ET chapter 9.1, 9.2.

Week 6, November 2: Vector Autoregressions and Factor models (cont.)

- Vector Auto Regressions (VARs)
  - Classical vs. Bayesian methods
  - Cointegration and error correction models
  - Spurious correlation
- Forecasting with very large data sets
  - Factor models
  - Panel data
  - Real-time data
- Spatial forecasting

Assignment 2 due

Readings:

ET chapter 9.3, 9.6, 10.

Background readings

Ghezzi, F., A. Timmermann, and M. Yang, 2026, The Hourly Economy: Measuring Local Economic Activity at High Frequency. UCSD working paper.

Karlsson, S., 2013, Forecasting with Bayesian Vector Autoregressions. Chapter 15 (pages 791-898) in G. Elliott and A. Timmermann (eds.): Handbook of Economic Forecasting, vol. 2.

Stock, J.H., and M.W. Watson, 2006, Forecasting with Many Predictors. Chapter 10 (pages 515-554) in G. Elliott, C.W.J. Granger and A. Timmermann (eds.): Handbook of Economic Forecasting, vol. 1.

Week 7, November 9: Prediction Markets, LLM-Based Forecasting, and Event/Density/Volatility Forecasting

- Event Forecasting models
  - Linear probability model
  - Logit/probit

- Sign predictions
- Brier Score

- Prediction Markets
    - Growth of Kalshi/Polymarket-style markets into macro, election, and business-outcome contracts;
- LLMs as forecasters: judgmental/event forecasting vs. superforecasters (Metaculus, ForecastBench tournaments) (Karger et al, 2025)
- LLMs as measurement tools: text-to-signal econometrics, training-data leakage, and validation-sample requirements for valid inference (Ludwig et al, 2025)
- Interval and quantile forecasts
- Density and volatility forecasting

Readings:

ET chapter 13

Bartlett, R. and M. O'Hara, 2025, Adverse selection in prediction markets: Evidence from Kalshi. Unpublished working paper, Stanford and Cornell.

Diercks, A.M., J. D. Katz, and J. W Wright, 2026, Kalshi and the Rise of Macro Markets. Federal Reserve Board, Washington DC.

Gomez Cram, R., Y. Guo, T. I. Jensen, and H. Kung, 2025, Financial Prediction Markets: A new Measure of Earnings Expectations. SSRN paper.

New background readings on LLM-based forecasting:

Ludwig, J., S. Mullainathan, and A. Rambachan, 2025, Large Language Models: An Applied Econometric Framework. Forthcoming, Annual Review of Economics, 2026.

Karger, E., H. Bastani, C. Yueh-Han, Z. Jacobs, F. Zhang, and P.E. Tetlock, 2025, ForecastBench: A Dynamic Benchmark of AI Forecasting Capabilities. arXiv:2409.19839.

Forecasting Research Institute / Good Judgment, 2025, Human vs. AI Forecasts (comparison of superforecaster and frontier-LLM difficulty-adjusted Brier scores).

Jadhav, A. and V. Mirza, 2025, Large Language Models in Equity Markets: Applications, Techniques, and Insights. Frontiers in Artificial Intelligence (survey of 84 studies, 2022–early 2025)

Week 8, November 16: Forecast Combination

- Forecast combination: A simple method that often beats the single best forecaster
- Why combine forecasts?
- Forecast encompassing – does one forecast dominate?
- Portfolios of forecasts
- Regression methods for forecast combination
    - Combining human, model, market, and LLM forecasts ("wisdom of the silicon crowd")

Assignment 3 due

Readings:

ET chapter 14.

Background readings

Diebold, F.X. and M. Shin, 2019, Machine learning for regularized survey forecast combination: Partially-egalitarian LASSO and its derivatives. International Journal of Forecasting 1679-1691.

Elliott, G., A. Gargano and A. Timmermann, 2013, Complete Subset Regressions. Journal of Econometrics 177, 357-373.

Makridakis S., E. Spiliotis, and V. Assimakopoulos, 2018, The M4 Competition: Results, findings, Conclusion, and Way Forward. International Journal of Forecasting 802-808.

Rapach, D., J.K. Strauss, and G. Zhou, 2010, Out-of-Sample Equity Premium Prediction: Combination Forecasts and Links to the Real Economy. Review of Financial Studies, 23(2), 821-862.

Week 9, November 23: Forecast evaluation and forecast comparison

- Forecast Evaluation
- Properties of optimal forecasts
- Regression tests
- Measures of forecast precision
- Comparing predictive accuracy
- Encompassing tests
- Look-ahead bias

Readings:

ET chapter 15.3, 17.1 – 17.4

Week 10, November 30: Model Breakdown. Pitfalls in forecasting. Data mining.

- Breaks in forecasting models
- Detection
- Importance for forecasting
- Skill or Luck? The Effect of Data Mining on Forecasting
- Data mining defined
- Newsletter, lucky penny examples
- In-sample versus out-of-sample forecasts (recap)
- Cross-validation
- Methods for handling data mining
    - Memorization and training-data leakage as a data-mining-like pitfall specific to LLM-based forecasts

Readings:

ET chapter 17.5-17.10, 19

Background readings

McLean, R.D., and J. Pontiff, 2016, Does Academic Research Destroy Stock Return Predictability? Journal of Finance 71, 5-32.

Week 11: Final Project is Due on Monday, December 7 at 8 pm.